

Response to Referee Report

Manuscript: "High-power continuous-wave optical waveguiding in a semiconductor nanowire"

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Dear Editor,

Thank you very much for sending us the referee's comments on our manuscript entitled "High-power continuous-wave optical waveguiding in a semiconductor nanowire." We have carefully considered all the comments and suggestions, and provide our point-by-point responses below. In the revised manuscript, all changes are highlighted in red.

Along with the clean version of the revised manuscript, a version with revision marks is also provided.

We hope the revised manuscript is now acceptable for publication in Advanced Photonics Research.

Sincerely yours,

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Referee's Comments and Our Responses

1. In Figure 2c, the output power increases strictly linearly with input power up to 2.5 W of guided power, with no sign of thermal-induced deviation from linearity. For a subwavelength-diameter wire suspended in air, I would have expected some deviation from linearity at these power levels due to absorption-induced heating. Can the authors comment on what ensures that the thermal equilibrium is maintained throughout the entire tested range?

Response: We appreciate the referee's attention to this point. Indeed, the strict linearity is one of the key observations of this work. The underlying physics can be understood from the balance between heat generation and dissipation:

On the generation side, CdS at 1550 nm is far below its bandgap (~ 2.42 eV), and the residual absorption coefficient is extremely low. As a result, the rate of heat generation increases only slowly with guided power. On the dissipation side, the surface-to-volume ratio of a 550-nm-diameter wire is approximately $4/d \approx 7.3 \mu\text{m}^{-1}$, which is orders of magnitude larger than that of bulk geometries. This large surface area enables efficient heat transfer to the surrounding air.

The combination of minimal heat generation and efficient dissipation ensures that the steady-state temperature rise remains modest. The Varshni-effect measurement (Figure S4) confirms this, yielding a temperature of ~ 180 °C at 1.1 W guided power — well below the thermal damage threshold of CdS (>900 °C). Even at the maximum guided power of 2.5 W, the estimated temperature (obtained by linear extrapolation of the Varshni calibration) remains well within the safe operating range. The absence of hysteresis in the round-trip power sweeps (Figure S3) further confirms that the thermal equilibrium is fully reversible.

In the revised manuscript, we have added a brief discussion of this thermal balance in Section 2.2.

2. The PL spectra in Figure 1e show a striking contrast between the CdS and ZnO nanowires: CdS exhibits only clean band-edge emission, while ZnO shows a prominent defect band around 550 nm attributed to oxygen vacancies. Meanwhile, the waveguiding performance in Figures 2c and 2d differs by orders of magnitude between the two materials. Is the defect absorption in ZnO, particularly at 532 nm where the defect states are absorptive, the primary factor that limits its guided power? I think it would be natural to connect these observations more explicitly.

Response: We thank the referee for identifying this connection. The referee is correct that the defect-state properties revealed by PL spectroscopy provide a direct physical explanation for the performance gap.

Defect states within the bandgap act as sub-bandgap absorption centers that convert guided optical power into heat. Under CW illumination, this heat accumulates continuously — unlike pulsed excitation, where the interpulse intervals provide cooling windows. Consequently, the power-handling capacity under CW conditions is especially sensitive to the defect density. The oxygen-vacancy-related band observed in ZnO (Figure 1e, ~ 550 nm) is a direct spectral signature of such absorbing centers, and its spectral position overlaps with the 532 nm waveguiding wavelength, where these defect states

exhibit significant absorption cross-sections. This accounts for the relatively low guided power of 33 mW in ZnO at 532 nm.

In contrast, the CdS nanowire shows no detectable defect luminescence, indicating a negligibly low density of sub-bandgap absorbing centers. At 1550 nm, which is far below the CdS bandgap, the defect absorption is essentially absent, allowing heat generation and dissipation to remain in balance and supporting the linear, hysteresis-free power transmission shown in Figure 2c.

We have added this discussion to Section 2.2 in the revised manuscript.

3. The SHG and THG conversion efficiencies (6.9×10^{-6} and 6.4×10^{-8}) are described as comparable to values obtained under pulsed excitation. I find this somewhat surprising — in pulsed excitation, the peak power density (typically $\sim \text{GW}/\text{cm}^2$) drives the nonlinear process, whereas under CW operation the power density is orders of magnitude lower. Could the authors comment on what physical mechanism enables this comparability?

Response: We thank the referee for this question and take this opportunity to clarify a point that may also be of interest to other readers. The premise of the question — that the CW power density is orders of magnitude lower than pulsed peak values — deserves closer examination. In our experiment, the guided power density at 1.1 W in a 550-nm-diameter CdS nanowire reaches $1.1 \times 10^{13} \text{ W}/\text{m}^2$ ($\sim 1.1 \text{ GW}/\text{cm}^2$). For comparison, the peak power densities reported in the pulsed nanowire studies cited in our manuscript are: $\sim 1 \text{ GW}/\text{cm}^2$ for Ag-coated CdS nanowire SHG [36], $\sim 10\text{--}100 \text{ GW}/\text{cm}^2$ for Si nanoplasmonic waveguide THG [37], and $\sim 0.6\text{--}29 \text{ GW}/\text{cm}^2$ for GaAs nanowire SHG [38]. The CW power density achieved here is therefore within the same order of magnitude as the peak power densities of refs. [36] and [38], and about one to two orders below ref. [37].

Given that the nonlinear conversion efficiency scales with power density, the comparable efficiencies are therefore not unexpected — the CW power density inside the nanowire overlaps with the range of peak power densities used in pulsed nanowire studies. This is a direct consequence of confining watt-level CW power into a subwavelength cross-section. The distinction lies not in the instantaneous nonlinear interaction strength — which is comparable — but in the temporal nature of the output: CW operation provides a continuous, stable harmonic signal, rather than the intermittent output characteristic of pulsed excitation.

The observed power-law slopes in the log-log plot (Figure 4d) — 1.97 ± 0.14 for SHG and 3.2 ± 0.18 for THG — are consistent with the expected quadratic and cubic dependencies, confirming that the nonlinear processes under CW excitation follow the same fundamental scaling laws as under pulsed excitation.

We have added this clarification to Section 2.4 in the revised manuscript.

4. The conclusion states that the results "may push semiconductor nanowire photonics into the high-power regime and open new opportunities." The demonstrations here are limited to CdS and ZnO at specific wavelengths. Could the authors comment on the generality of their approach — for instance, what constraints need to be met for other semiconductor nanowire materials?

Response: We agree that the conclusion should better reflect the scope of the current work. In the revised manuscript, we have rewritten the conclusion as follows:

"In summary, we have demonstrated high-power CW optical waveguiding in CdS and ZnO nanowires. A single 550-nm-diameter CdS nanowire safely waveguides 1550-nm CW light with power up to 2.5 W, nearly four orders of magnitude higher than previously reported results; a 690-nm-diameter ZnO nanowire supports guided powers of 292 mW at 1550 nm and 33 mW at 532 nm. These results were enabled by the interplay of low-defect nanowires and efficient fiber-taper evanescent coupling. Building on this high-power platform, we further realized intermodal-phase-matched CW second- and third-harmonic generation with efficiencies comparable to pulsed excitation. While the current demonstrations are limited to CdS and ZnO at specific wavelengths, the fiber-taper coupling approach is in principle applicable to other semiconductor nanowire materials, provided that three conditions are simultaneously met: low defect density, operation in a weak-absorption spectral window, and effective index matching for efficient coupling. Extending this approach to a broader range of materials and wavelengths may open new opportunities in nonlinear optics, sensing, and integrated photonic systems."

Minor Comments

5. The Varshni fitting parameters in Equation (3) ($\alpha = 0.00059$, $\beta = 33.5$) are given without units.

Response: We have added the units in the revised manuscript: $\alpha = 5.9 \times 10^{-4}$ eV/K, $\beta = 33.5$ K, and $a_0 = 2.459$ eV.

6. The Supporting Information is referenced in the main text via figure citations (e.g., Figure S2), but direct references to the corresponding sections (S1–S5) would make it easier for readers to locate the relevant discussions.

Response: We have updated the cross-references in the main text to include the section numbers: surface roughness measurement (see Supporting Information S1), coupling efficiency simulation and waveguiding power calculation (see Supporting Information S2 and S3), temperature evaluation via the Varshni effect (see Supporting Information S4), and mode analysis (see Supporting Information S5).

List of Changes in the Revised Manuscript

Main text:

1. In Section 2.2, a discussion of the thermal balance underlying the linear Pout–Pin relationship has been added.
2. In Section 2.2, a discussion connecting the PL defect characterization (Figure 1e) to the CdS–ZnO waveguiding performance gap has been added.
3. In Section 2.4, a quantitative comparison of the CW power density ($\sim 1.1 \text{ GW/cm}^2$) with pulsed peak power densities from refs. [36–38] ($\sim 0.6\text{--}100 \text{ GW/cm}^2$) has been added, clarifying the origin of the comparable nonlinear conversion efficiencies.
4. The conclusion has been revised to acknowledge the material- and wavelength-specific conditions while noting the three requirements for extending the approach.
5. The units of the Varshni parameters in Equation (3) have been added: $\alpha = 5.9 \times 10^{-4} \text{ eV/K}$, $\beta = 33.5 \text{ K}$, $a_0 = 2.459 \text{ eV}$.
6. Cross-references to Supporting Information sections S1–S5 have been added in the main text.